

Analyzing E-Learning Platform Purchases Using MySQL

Module-End Assignment 3 - One-Page Summary Report

Project Objective

The objective of this project was to design and analyze an e-learning purchase database using MySQL. The analysis focused on sales trends, learner purchasing behavior, course popularity, category performance, and revenue.

Database & Analysis

The database contains three related tables: learners, courses, and purchases. Primary keys and foreign keys were used to maintain relationships. The project applied INNER JOIN, LEFT JOIN, RIGHT JOIN, aggregate functions, GROUP BY, HAVING, subqueries, correlated subqueries, a CTE, CASE expression, NULL handling, and a SQL view.

Key Insights

- The Advanced category generated the highest revenue in the sample dataset, supported by Advanced SQL and Data Analytics Masterclass purchases.
- Data Analytics Masterclass produced the largest single purchase amount: Rs. 24,000 (2 courses x Rs. 12,000).
- Python for Beginners was purchased in high quantity, including a 3-course purchase worth Rs. 15,000.
- Learners purchasing across multiple categories can be identified for cross-selling and personalized course recommendations.
- The analysis also identifies above-average spenders and classifies learners as High Value, Medium Value, or Low Value based on total spending.
- NULL handling ensures courses with no purchases can still be reported with a purchase count of 0.

Recommendations

- Focus marketing campaigns on high-revenue and high-demand course categories.
- Target high-value learners with advanced courses, bundles, and personalized recommendations.
- Promote less-purchased courses through discounts, introductory offers, or course bundles.
- Use category and learner spending analysis regularly to improve pricing and marketing decisions.

Conclusion

This project demonstrates how MySQL can transform relational purchase data into useful business insights. The queries provide a clear view of learner spending, course demand, category revenue, and customer value, helping the e-learning platform make better data-driven decisions.